

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section:

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

Source CSV files are available at the specified locations:

``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and
``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``

- Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and load into a Delta table named "customer" with schema: CustId, Name, EmailId, Region.
2. Read order CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`` and load into a Delta table named "order" with schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.
3. Add a new column "TotalAmount" to the "order" table, calculated as the product of "PricePerUnit" and "Qty".

FR-DLT-002: Clean and Transform Customer and Order Data

Title: Clean and Transform Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order Delta tables.

Preconditions:

- Customer and order Delta tables are loaded with data

Main Flow / Functional Steps:

1. Remove null records from the "customer" and "order" Delta tables.
2. Remove duplicate records from the "customer" and "order" Delta tables.

FR-DLT-003: Create and Load Order Summary Table

Title: Create and Load Order Summary Table

Description: The system must create an "ordersummary" table if it does not exist and load it with joined customer and order data using SCD type 2.

Preconditions:

- Customer and order Delta tables are cleaned and transformed
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "ordersummary" table if it does not exist in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" with schema: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date.
2. Join the "customer" and "order" Delta tables on "CustId" and load the resulting data into the "ordersummary" table using SCD type 2.

FR-DLT-004: Update Order Summary Table on Customer Changes

Title: Update Order Summary Table on Customer Changes

Description: The system must update the "ordersummary" table when there are changes to the "customer" table.

Preconditions:

- "ordersummary" table is loaded with data
- Changes are made to the "customer" table

Main Flow / Functional Steps:

1. Identify changes to the "customer" table.
2. Update the "ordersummary" table by making old records inactive and new records active, and updating StartDate and EndDate accordingly.

FR-DLT-005: Create and Load Customer Aggregate Spend Table

Title: Create and Load Customer Aggregate Spend Table

Description: The system must create a "customeraggregatespend" table if it does not exist and load it with aggregated data from the "ordersummary" table.

Preconditions:

- "ordersummary" table is loaded with data

- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table if it does not exist in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" with schema: Name, TotalAmount, Date.
2. Aggregate the "TotalAmount" column from the "ordersummary" table by "Name" and "Date" and load the resulting data into the "customeraggregatespend" table.

FR-DLT-006: Implement Delta Live Tables**Title: Implement Delta Live Tables**

Description: The system must implement Delta Live Tables to execute the above requirements.

Preconditions:

- All previous requirements are implemented

Main Flow / Functional Steps:

1. Implement Delta Live Tables to execute the data pipeline from customer and order data loading to customer aggregate spend data loading.